

# NEERAJ KUMAR SINGH

## Contact

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## Profiles

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<https://nrjsingh1.github.io/>

## Summary

Senior Software Engineer with 9 years of enterprise experience designing high-throughput microservices, scalable distributed systems, and event-driven architectures. Currently pursuing an MS in Computer Science. Seeking a Backend Engineering Co-op role starting Mid-September 2026 (fully eligible for university-authorized CPT work initialization)

## Skills

JAVA, SPRING, POSTGRESQL, HIBERNATE, KAFKA, SPARK, WEB DEVELOPMENT, REST, MICROSERVICES  
PYTHON, KUBERNETES, AWS, GIT, JENKINS, MAVEN, JIRA, AGILE

## Experience

**Here Technologies, Bengaluru** , Senior Software Engineer Sept 2021 - May 2025

### Spark Based Road Attribute Automation

- Migrated road attribute automation application from an event-stream-based processing system to a Spark-based batch processing job.
- Optimized data pipeline throughput, reducing batch processing execution latency by 83% (from 24h to 4h) by implementing a distributed MapReduce structure on AWS EMR clusters.
- Integrated the batch job into the Directed Acyclic Graph (DAG) execution plan, ensuring coordination with upstream and downstream processes for successful execution.

### Event Stream-Based Processing

- Developed an event-driven streaming application using Java, Spring Boot, PostgreSQL, and Apache Kafka to process real-time data efficiently.
- Implemented a blocking queue mechanism to temporarily store incoming events from upstream systems and persist them in an RDBMS for intermediate storage.
- Designed and scheduled event processing workflows to handle events sequentially and publish processed data to downstream consumers.

**Cisco India, Chennai** , Software Engineer II Feb 2020 - Sept 2021

### Cisco Cloud Calling

- Designed and implemented RESTful and SOAP-based APIs using Spring Boot and PostgreSQL, ensuring efficient and scalable backend services.
- Integrated REST clients with an Angular-based front-end, facilitating seamless communication between back-end and UI components.
- Developed a Spring-based authentication service to enhance security and user management.
- Automated UI testing using Selenium, improving application reliability and reducing manual testing efforts.

**Verizon India, Chennai** , Analyst System Developer July 2019 - Feb 2020

### Conversation Manager Tool

- Worked on the backend of a SaaS product to design conversation flows, using Java PostgreSQL RESTful services.

**Verizon India, Chennai** , Software Engineer Developer July 2016 - July 2019

### Wireline Voice Portal - IVR

- Worked as Java developer in Verizon's Voice portal team, focused on enhancing DIY/ self-service and reducing call misdirects.

- Developed menu and DTMF based IVR call flows to provide self-service using VXML (Voice XML) and Nuance grammars.

#### Angular JS based Metrics Dashboard

- Created Voice Portal Metrics Dashboard, showing the real-time call volumes for different kinds of transaction.

#### Security Vulnerability Finder Application:

- Java based text scanning tool to scan the application logs to find CPNI based customers sensitive data and mask them before sending to third parties.

#### Verizon Store Locator App:

- Created webapp to find the nearest Verizon store from the customer location , using Php , CSS , HTML and Google map API.

### Computer Science Department, CUSAT , [Internship](#)

[July 2015 - April 2016](#)

#### Computer Aided Process Planning, HMT Kochi

- Developed Computer Aided Process Planning python tool to read CAD diagrams of machine tools , using dxfgrabber library
- used RDBMs to store mapping of basic shapes with their planning steps.
- Generated sequence of processes required in manufacturing of machine tools
- Worked in collaboration with Hindustan Machine Tools (HMT Kochi)

#### College Website, School Of Engineering, Kochi

- Developed official website for college - [soe.cusat.ac.in](http://soe.cusat.ac.in) , using PHP, HTML , CSS, JQuery , MySQL, Tom-cat.
- Publish events, news and updates as updated by Admins
- Created profiles for Professors showing published papers and researches. Login and Authentication for all users.

## Education

### Montclair State University, New Jersey, USA ,

[2025 - 2027](#)

#### Master of Computer Science

### Cochin University of Science & Technology, Kerala, India ,

[2012 - 2016](#)

#### Bachelor of Technology, Computer Science Engineering

### St Michael's High School, Patna, Bihar, India ,

[2009 - 2011](#)

#### All India Senior School Certificate Examination (AISSCE) (12th), CBSE

### Mother's International Academy, Patna, Bihar, India ,

[2009](#)

#### All India Secondary School Examination (10th), CBSE